MGU-BCA (HONOURS) REGULAR XAMINATION March -2026

Fourth Semester

Skill Enhancement Course (SEC) - MG4SECBCA201-PROBABILITY DISTRIBUTIONS AND STATISTICAL INFERENCE

(2024 ADMISSION ONWARDS)

Scheme of Valuation

Part A

Answer any 5 questions, each question carries 2 marks

1. k= 1/6 – give 2 marks.
2. Any one assumption of the chi square test - give 2 marks.
   1. The observations are independent of each other
   2. The data are in the form of counts
   3. The expected frequency in each cell should generally be greater thn 5
   4. The total number of observations fixed.
3. P[Z < 2.34]= 0.9904 –give 2 marks.
4. D.f= 25 –give 2 marks.
5. Chi-square distribution- 2 marks.
6. In practice, one chooses α (commonly 0.10, 0.05 or 0.01) before looking at the data. If the test's p-value is less than α, we reject the null hypothesis and the result is called statistically significant – 2 marks.
7. A Standard Normal Distribution is defined as a special form of the normal distribution where the mean (average) is 0 and the standard deviation is 1. –give 2 marks.
8. $Z = \frac{P-p_0}{\sqrt{\frac{p_0 q_0}{n}}} \sim N(0,1)$- give 2 marks.

Part B

Answer any 5 questions, each question carries 6 marks

9. Definition of mathematical expectation either discrete case or both -3 marks any 3 properties-3 marks
10. Definition of Bernoulli distribution – 1 ½ marks mean -1 ½ marks variance -1 ½ marks mgf -1 ½ marks
11. Form of F distribution -4 marks properties and uses – 2marks
12. Hypothesis $H_0 : \mu = 5$ Vs $H_1 : \mu \neq 5$ - 1 ½ marks

    test statistic $Z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$- 1 ½ marks

    Z= -3.37 Z_α= 2.58 – 2 marks Reject H_0 the net weight is not 5 kg – 1 marks.
13. The value of the parameter λ=1 -4 marks P[X=0] = 1/e= 0.3679 -2 marks.
14. Fitting of binomial distribution -6 marks (either theoretical or by using an example).

15. Any three differences – give 6 marks.
16. Definition of null and alternative – 4 marks examples -2 marks.

Part C

Answer any 2 questions, each question carries 15 marks.

17. Definition of random variable with example- 4 marks Different types of random variables with examples -6 marks real life applications, examples and characteristics- 5 marks.
18. Normal distribution – 8 marks any 5 properties -7 marks.
19. Difference between statistic and parameter – 8 marks.
    Difference between standard deviation and standard error – 7 marks.
20. Hypothesis H_0 :Mendes's observed data fits the theoretical ratio Vs H_1 :doesnot fit - 2 marks.
    Test Statistic $\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i} \sim \chi^2$ distribution with (n-1) degrees of freedom -2 marks.
    Expected frequencies 336, 112, 74.67 an 37.33 -5 marks.
    $\chi^2 = 17.32$ - 2 marks $\chi^2{}_{\propto} = 7.815$ d.f= n-1=3 – 2 marks.
    Reject the null hypothesis.
    There is significant difference between observed and expected frequencies. Hence the experimental results do not fit Mendel's theoretical ratio. -2 marks.

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